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VIVEKANAND EDUCATION SOCIETY
INSTITUTE OF TECHNOLOGY
(AUTONOMOUS)

Department of Computer Engineering

CAPSTONE PROJECT LOGBOOK

(26CMCP71 : Capstone Project - II)

2026-27

GROUP MEMBERS

Sakshi Thorat (D17B/55)

Diksha Patkar (D17C/42)

Pradnya Patil (D17C/39)

Akritee Singh (D17B/69)

Project Mentor

Dr. Rohini Temkar

Assistant Professor, Department of Computer Engineering

VIVEKANAND EDUCATION SOCIETY
INSTITUTE OF TECHNOLOGY (AUTONOMOUS)

Department of Computer Engineering

INSTITUTE VISION & MISSION

VISION:

To build a vibrant, knowledge-driven ecosystem that promotes holistic learning, inspires innovation, practices sustainability and value-based education, and shapes socially conscious, industry-ready technocrats.

MISSION:

- To inculcate a culture of multidisciplinary value based education.
- To inspire students to apply their knowledge to meet the needs of industry and society.
- To foster research, entrepreneurship and lifelong learning for societal growth

COMPUTER ENGINEERING DEPARTMENT

VISION:

To establish a center of excellence in computing by empowering students with advanced technological skills for fostering innovation, research, and societal impact.

MISSION:

- To cultivate a teaching learning environment that unites theory with computing skills, empowering students to adapt to evolving technologies.
- To strengthen core computing knowledge by fostering research, innovation, and entrepreneurship through multi-disciplinary learning.
- To prepare competent professionals with industry-ready skills, encouraging social awareness, sustainable development, and a commitment to lifelong learning.

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PROGRAM EDUCATIONAL OBJECTIVES (PEO's)

I	To equip Computer Engineering students with a strong foundation in mathematics, science, and core computing and engineering principles, enabling them to comprehend, analyze, and design effective technology-driven solutions.
II	To foster a multidisciplinary mindset in Computer Engineering students through flexible learning, enabling adaptability along with strong communication, teamwork, and leadership skills.
III	To empower Computer Engineering students with advanced computing and software skills, fostering innovation, entrepreneurship, and sustainable practices for lifelong learning.
IV	To provide Computer Engineering students with broad educational and research experiences through academic and industrial collaboration programs, preparing them for industry-ready careers.

PROGRAM OUTCOMES (POs)

Program Outcome Code	Program Outcome Description
PO1	**Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop the solution of complex engineering problems.
PO2	**Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
PO3	**Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
PO4	**Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.(WK8).

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PO5	**Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
PO6	**The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.(WK1, WK5, and WK7).
PO7	**Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
PO8	**Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
PO9	**Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural,language, and learning differences.
PO10	**Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
PO11	**Life-Long Learning:** Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1	**Professional Skills -** The ability to develop programs for computer based systems of varying complexity and domains using standard practices.
PSO2	**Successful Career -** The ability to adopt skills, languages, environment and platforms for creating innovative career paths, being successful entrepreneurs or for pursuing higher studies.

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STUDENT INFORMATION

Project Title: A Unified AI And Blockchain Framework For Jasmine Disease Detection, Yield Prediction, Smart Irrigation, Harvest Planning, And Traceability

SDGs Mapped: SDG 9, SDG 12, SDG 13

	Student 1	Student 2	Student 3	Student 4
Roll No.	55	42	39	69
Name	Sakshi Thorat	Diksha Patkar	Pradnya Patil	Akritee Singh
Class with Division	D17B	D17C	D17C	D17B
Contact No.	8591764641	8591553536	7756954142	8425013620
E-mail	2023.sakshi.thorat@ves.ac.in	2023.diksha.patkar@ves.ac.in	2023.pradnya.patil@ves.ac.in	d2023.akritee.singh@ves.ac.in
Address	D-2/203, Sahakar Darshan	1/6, Vishwa Roop Darshan CHS	Nad Sangeet Vidyalaya	D/901, Sai Radha Complex
	Annapurna Nagar	Durga Mata Mandir Road	Near Cake Shop, Behind Borba Devi Mandir	L.B.S Marg, Near Asian Paints
	Adharwadi	Kolsewadi	Govandi - East,	Bhandup (West)
	Kalyan(W), 421301	Kalyan(E), 421306	Mumbai, 400088	Mumbai-400078

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INSTRUCTIONS TO STUDENTS:

1. The logbook must be submitted to the Project Mentor or Project Coordinator for verification and evaluation of project activities at least once in a week.
2. A logbook duly signed by the project mentor must be submitted with a project blackbook for evaluation at the end of semester to the department.

DECLARATION

I declare that this project represents my ideas in my own words and wherever others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my project work. I promise to maintain minimum 75% attendance, as per the Institute norms. I understand that any violation of the above will be cause for disciplinary action by the Institute.

Yours Faithfully

Sakshi Thorat (D17B/55)

Diksha Patkar (D17C/42)

Pradnya Patil (D17C/39)

Akritee Singh (D17B/69)

(Signature of Students)

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Letter of Acceptance

I undersigned, **Dr. Rohini Temkar** working in the Computer Engineering department, willing to guide the Project titled ***A Unified AI And Blockchain Framework For Jasmine Disease Detection, Yield Prediction, Smart Irrigation, Harvest Planning, And Traceability*** for the Capstone Project - II Semester VII respectively for the ***Academic Year 2026-27.***

The names of the students are:

1. **Sakshi Jotiram Thorat (D17B-55)**
2. **Diksha Vinit Patkar (D17C-42)**
3. **Pradnya Prabhakar Patil (D17C-39)**
4. **Akritee Singh (D17B-69)**

[redacted]	[redacted]	[redacted]
(Project Mentor)	(Capstone Project Coordinator)	(HOD Department of Computer Engineering)

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COURSE OUTCOMES FOR CAPSTONE PROJECT–II

Course Outcome	Description	PO's Covered	PSO's Covered
CO1	Implement and integrate all modules of the proposed system effectively.	PO1, PO2, PO3, PO5	PSO1, PSO2
CO2	Perform testing, debugging, validation, and performance optimization of the developed solution.	PO2, PO3, PO4, PO5	PSO1, PSO2
CO3	Analyze project outcomes using appropriate engineering tools and metrics.	PO2, PO4, PO5	PSO1, PSO2
CO4	Demonstrate deployment readiness, usability, security, scalability, and sustainability considerations.	PO3, PO5, PO6, PO7	PSO1, PSO2
CO5	Analyze, interpret, and present research outcomes effectively using appropriate research methodologies, analytical tools, and presentation techniques.	PO2, PO4, PO5, PO8, PO10	PSO1, PSO2
CO6	Present and defend the complete project professionally through demonstrations, presentations, and comprehensive technical documentation.	PO8, PO9, PO10, PO11	PSO1, PSO2

CO–PO–PSO Mapping Table

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	3	3	–	3	–	–	–	3	–	2	3	3
CO2	2	3	2	3	3	–	–	–	2	–	2	3	2
CO3	2	3	2	3	3	–	–	–	–	2	2	3	2
CO4	2	2	3	2	3	2	3	2	–	–	3	3	3
CO5	2	3	–	3	3	–	–	2	–	3	2	2	2
CO6	–	–	–	–	2	–	–	2	3	3	2	2	2

Mapping scale: 3 = High correlation; 2 = Moderate correlation; 1 = Low correlation; – = No direct correlation.

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SCHEDULE FOR CAPSTONE PROJECT - II

Date	Week	Contents	Remark	Guide Sign
30 Jul 2026	1	Reviewed Capstone I outcomes (EfficientNet-B0 disease classifier, Figma UI, client requirements). Defined the Semester VII scope: severity estimation, advisory, yield prediction, smart irrigation, harvest planning, blockchain traceability and system integration.	Scope and work plan finalized.	
06 Aug 2026	2	Studied literature and patents for the new modules only (yield prediction, irrigation scheduling, market-aware harvest planning, blockchain traceability). Literature for new modules completed	Literature for new modules completed.	
13 Aug 2026	3	Designed the extended system architecture, module interfaces, API contracts, database schema and user roles (Farmer, Expert, Admin).	Architecture approved by mentor.	
20 Aug 2026	4	Set up the FastAPI backend, MongoDB, Cloudinary and Firebase authentication. Deployed the trained EfficientNet-B0 model as a prediction API with image validation.	Backend and disease API ready.	
03 Sep 2026	5	Implemented severity estimation from the affected leaf-area ratio. Built the treatment advisory (chemical and organic options, dosage, safety notes) and the expert validation workflow with a confidence threshold.	Severity and advisory modules completed.	

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10 Sep 2026	6	Built the yield prediction module on the 20,000-record, 16-feature dataset. Compared Random Forest, XGBoost and SVR; XGBoost performed best ($R^2 = 0.8546$).	Yield module completed.	
24 Sep 2026	7	Implemented the smart irrigation module: irrigation-need classification and water-quantity estimation using XGBoost, with weather inputs from the OpenWeather API.	Irrigation module completed.	
01 Oct 2026	8	Developed the market-aware harvest module (demand, price, buyer acceptance, unsold risk) and the Harvest Suitability Score for ranking harvest dates.	Harvest module completed.	
08 Oct 2026	9	Implemented blockchain traceability with SHA-256 hash-linked blocks, unique Batch IDs, QR codes and role-based access control. Recorded 7 traceability events for a demonstration batch.	Traceability module completed.	
15 Oct 2026	10	Developed the Flutter farmer app (five languages) and the React admin and expert dashboards. Integrated all modules end to end.	Integrated prototype ready.	
29 Oct 2026	11	Performed unit, integration, input-validation and access-control testing and debugging. Analyzed performance with accuracy, F1, MAE, RMSE, R^2 and confusion matrices, and compared results with existing work.	Testing and analysis completed.	
05 Nov 2026	12	Wrote and refined the research paper and blackbook, ran plagiarism and AI-content checks, and prepared the final demonstration and presentation.	Ready for final submission .	

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PROGRESS / ATTENDANCE REPORT

Title of the Project: A Unified AI And Blockchain Framework For Jasmine Disease Detection, Yield Prediction, Smart Irrigation, Harvest Planning, And Traceability	
Group No. 17	**Sakshi Jotiram Thorat (D17B-55)** **Diksha Vinit Patkar (D17C-42)** **Pradnya Prabhakar Patil (D17C-39)** **Akritee Singh (D17B-69)**
Name of the Project Mentor: Dr. Rohini Temkar	

Sr. No	Date	Attendance				Progress/Suggestion	Mapping		
		1	2	3	4		CO	PO	PSO
1	30 Jul 2026					Reviewed Capstone I outcomes and defined Semester VII scope and plan.	CO1	PO1, PO2, PO3	PSO1, PSO2
2	06 Aug 2026					Completed literature and patent study for yield, irrigation, harvest and traceability modules.	CO5	PO2, PO4	PSO1, PSO2
3	13 Aug 2026					Finalized architecture, API and roles. The mentor approved the design.	CO1	PO1, PO3, PO5	PSO1, PSO2
4	20 Aug 2026					Set up the FastAPI backend. The mentor advised adding image validation.	CO1	PO3, PO5	PSO1, PSO2
5	03 Sep 2026					Implemented severity estimation, advisory and expert validation. The mentor advised routing advanced plan to experts.	CO1	PO2, PO3, PO5	PSO1, PSO2

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6	10 Sep 2026					Completed yield prediction; Mentor advised reporting MAE, RMSE and MAPE together.	CO1	PO3, PO4, PO5	PSO1, PSO2
7	24 Sep 2026					Completed the smart irrigation module with weather integration.	CO1	PO3, PO5	PSO1, PSO2
8	01 Oct 2026					Completed the harvest module using synthetic data.	CO3	PO2, PO4, PO5	PSO1, PSO2
9	08 Oct 2026					Implemented blockchain traceability with QR codes and role-based access. The mentor suggested adding tamper checks.	CO4	PO3, PO5, PO6, PO7	PSO1, PSO2
10	15 Oct 2026					Built the mobile app by integrated all modules.	CO1	PO1, PO3, PO5	PSO1, PSO2
11	29 Oct 2026					Completed testing, debugging and result analysis. The mentor advised marking planned tests separately from passed tests.	CO2	PO2, PO3, PO4, PO5	PSO1, PSO2
12	05 Nov 2026					Finalized the paper and blackbook and prepared the demo and presentation.	CO6	PO8, PO9, PO10, PO11	PSO1, PSO2

Sign of the Project Mentor

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EXAMINER'S FEEDBACK FORM

Name of External examiner:_____

College of External examiner:_____

Name of Internal examiner:_____

Date of Examination:_____/____/_____

No. of students in project team: 4 members

Availability of separate lab for the project: Yes / No

Student Performance Analysis (Put Tick as per your Observation)

Excellent (3) Very Good (2) Good (1)				
Sr. No.	**Observation**	**(3)**	**(2)**	**(1)**
1	Quality of problem and Complex Engineering Problems			
2	Innovativeness in solutions			
3	Cost effectiveness and Societal impact			
4	Full functioning of working model as per stated requirements			
5	Effective use of skill sets			
6	Effective use of standard engineering norms			
7	Contribution of an individual as member or leader			
8	Clarity in written and oral communication			
9	Overall performance			

Suggest new Innovative Technique/Idea/ objectives related to this project.

Signature of External Examiner **Signature of Internal Examiner**